

# VISHNUVARDHAN REDDY GANJI

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## EXPERIENCE

### FactSet

Jun 2023 – Present

#### Software Engineer

- Architected an **AI-powered summarization pipeline** integrating **16 data systems** that condenses market news into structured synopses across **12 sectors and 5 global regions** processing **2,000 articles per day**.
- Modernized production systems by upgrading **Node.js and AWS Lambda to Node 20**, migrating from **CentOS 7 to Rocky Linux 9**, and using **Terraform** to improve stability and maintainability.
- Improved reliability and operational response for Market Intelligence workflows by introducing **message-queue architecture**, optimizing **data-fetch concurrency**, and implementing **PagerDuty alerts** for disrupted jobs.
- Built **Bond Comps**, a full-stack **bond benchmarking platform** with a **Highcharts scatter visualization** and a bond matching engine spanning **1.5M+ bonds**, reducing relative-value analysis time by **70% for 200+ analysts**.
- Engineered **Capital Market Dashboard** (Vue 3/TypeScript/Express), a monitoring platform for **1,000+ analysts/traders** tracking **20,000+ securities**, with custom alerts and news, cutting deal-screening time by **50%**.

### FactSet

Jan 2023 – May 2023

#### Software Engineer Intern

- Built a **full-stack email readership analytics dashboard** delivering **content engagement visibility** across **40,000 emails**, directly enabling identification and deactivation of **1,800 non-read email subscriptions**.
- Designed an **asynchronous data processing pipeline** using Redis job queues to process **2.5M email sent logs and 1.2M read logs** per quarter from Elasticsearch, persisting results to database and cache.
- Engineered an **S3-backed CSV report generation pipeline** that processes data across a **configurable 90-day range**, generates Excel exports, and automatically delivers **24-hour expiring download links**.

## PROJECTS

### Chat with PDF   *chatwithpdf.pro* | *GitHub* | *Docs*

*Next.js, React, TypeScript, Convex, OpenAI, Mistral OCR, Clerk*

- Built a **full-stack RAG application** that lets users **chat with PDFs** and get answers **grounded strictly in the source document**, with inline citations.
- Engineered a **hybrid retrieval pipeline** with **1536-dimensional embeddings**, full-text search, and Reciprocal Rank Fusion, achieving **2s first token latency** and **15s average ingestion time**.
- Developed a **citation-validation pipeline** that verifies generated quotes against source text and resolves evidence to exact page spans achieving **92% citation accuracy** on a **50-question evaluation data set**.
- Architected a **retryable Convex ingestion workflow** for OCR, summarization, chunking, and embeddings, with persisted OCR checkpoints, **up to 3 processing attempts**, SSE streaming and per-user data isolation.

### Voice Flow   *voiceflow.app* | *GitHub* | *Docs*

*Electron, React, Zustand, TypeScript, Rust, Whisper API, Web Audio API*

- Built a **macOS menu-bar dictation app** that captures audio via global hotkeys and delivers **real-time voice-to-text transcription** using the Whisper API, with an optional **LLM-based polish pass** for cleanup.
- Engineered a **native Rust sidecar hotkey listener** integrated over IPC across Electron's main, renderer, and overlay processes, enabling recording independent of app focus.
- Implemented a **16 kHz PCM16 AudioWorklet pipeline** with **100 ms** chunks, adaptive **2-pass Whisper transcription** below a **0.70** quality threshold, and safe macOS Accessibility-based text injection.

## TECHNICAL SKILLS

**Core:** TypeScript, JavaScript, Python, C++, Rust, SQL

**Web Engineering:** React, Next.js, Vue.js, Node.js, Express.js, Tailwind CSS

**Systems & Data:** REST APIs, WebSockets, PostgreSQL, Authentication, Caching

**AI Engineering:** LLM APIs, RAG, Embeddings, OCR, Vector Search

**Infrastructure:** AWS, Terraform, Vercel, Linux, Git, GitHub Actions

## AWARDS / ACCOMPLISHMENTS

- Received **Blue Ribbon Award** twice for outstanding performance and high-impact contributions within the team.
- Achieved **LeetCode Guardian** with **2,200+ rating**, solving 1,500+ problems in data structures and algorithms.
- Led a team to the **finals of FactSet's internal hackathon**, selected from 250+ competing projects.

## EDUCATION

Gokaraju Rangaraju Institute of Engineering and Technology

Aug 2019 – May 2023

*Bachelor of Technology in Computer Science and Engineering*

*Hyderabad, Telangana*